Pharmacology of Sympathomimetics, Part 1
Direct-Acting Sympathomimetics

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I am available to groups and individuals for pharmacology help and discussions, including the Practice Questions, by appointment.

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After completion of the preparation materials, students should be able to:

1. Drug classes

Identify the three mechanistic categories of the sympathomimetic drugs.

2. Structure-activity relationships → actions, uptake, and pharmacokinetic properties

Correlate the structures of the natural catecholamines and the sympathomimetics with their actions at adrenergic sites, uptake by NET, and pharmacokinetic properties.

3. SNS physiology → pharmacologic effects

Relate the physiology of the sympathetic nervous system and reflex responses to the specific effects of each subclass of sympathomimetics.

4. Pharmacologic effects → therapeutic uses

Relate the specific effects to the therapeutic uses of each subclass of sympathomimetics.

5. Actions → adverse effects and cautions

Predict the adverse effects, drug interactions, and contraindications from the actions of the sympathomimetics on sympathetic nervous system functions.

Preparation Materials

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Required

- **ScholarRx Bricks | Practice Questions**
- Suggested supplemental resources if you wish to use them:
- Video Lecture | Dr. Goldstein's Notes Handout | Guided Reading Questions
- Links to pharmacology textbooks and review books are provided on the next slide.

SUGGESTIONS:

- **Use the resources that work best for you.**
- **You do not need to study all of them. (ScholarRx Bricks & Practice Questions are required.)**
- **Work through the GUIDED READING QUESTIONS with pen/pencil and paper.**

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- **Practice questions (not graded): Simple Recall and Case Vignettes**

Scholar Rx Bricks:

Name of Discipline/ Organ System: General Pharmacology > Autonomic Drugs > Drugs that Alter Sympathetic Tone

Title of Bricks: Sympathomimetics > Sympathomimetic Agents

Link(s): https://exchange.scholarrx.com/brick/sympathomimetic-agents

Recommended supplemental resources (not required)

Katzung's Basic & Clinical Pharmacology, 16e, 2024: Chapter 9: Adrenoceptor Agonists & Sympathomimetic Drugs

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281747810

The review books below provide practice questions.

LWW Health Library, Medical Education: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 6: Adrenergic Agonists (Review questions are included in the chapter.)

https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325673&bookid=3222

Katzung's Pharmacology Examination and Board Review, 14e, 2024: Chapter 9: Adrenoceptor Agonists & Sympathomimetic Drugs

https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3461&sectionid=285590222

Key points

- The core concepts of sympathomimetic pharmacology are built on knowledge of the physiological actions and reflex regulatory responses of the autonomic nervous system.
- ***Knowledge of the physiology of the autonomic nervous system is essential*** for understanding the pharmacology of the sympathomimetic agents and their clinical applications:
  - Adrenergic and cholinergic neurotransmitters,
  - G protein-coupled receptor (GPCR) receptor types and actions,
  - Effector responses to sympathetic and parasympathetic stimulation, and
  - Reflex regulatory responses (compensatory responses).

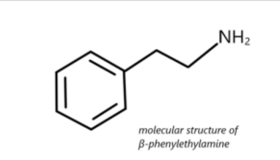
Key points:

Drug classes

The mechanistic classes of the sympathomimetics are:

- **Direct-acting** drugs bind to and activate adrenergic receptors producing a response
- **Indirect-acting** drugs augment the activity of the catecholamine
  - Drugs that induce catecholamine release from storage vesicles → increase neurotransmitter availability
  - Drugs that inhibit reuptake into adrenergic neurons→ increase synaptic concentrations of the catecholamine
  - Drugs the block degradation of catecholamines→ increase catecholamine concentrations
- **Mixed-acting**
  - Drugs that directly stimulate adrenergic receptors AND induce release from storage vesicles

Structure-activity relationship → actions, uptake, and pharmacokinetic properties



- Most adrenergic drugs are derivatives of β-phenylethylamine (right figure).
- Substitutions on the benzene ring or on the ethylamine side chains produce a variety of compounds with varying abilities to differentiate between α and β receptors and to penetrate the CNS.
- Two important structural features of these drugs are (1) the number and location of OH substitutions on the benzene ring and (2) the nature of the substituent on the amino nitrogen.
- The structural characteristics determine potency at the receptors, uptake by norepinephrine transporter, and pharmacokinetic properties, such as ability to penetrate into the CNS, metabolism by COMT and MAO, and half-life.

Key points

Physiology-pharmacology relationship

- The sympathetic nervous system 1) increases heart rate and blood pressure, 2) dilates airways, 3) dilates pupils, 4) mobilizes energy stores, and 5) redirects blood flow to the muscles by constricting peripheral and mesenteric vessels and simultaneously dilating the skeletal muscle vasculature.
- Sympathomimetic drugs mimic these actions.
- The pharmacologic effects are dependent on the mechanisms of each drug – nonspecific vs. specific actions on receptors, direct vs. indirect actions.

Pharmacologic effects → therapeutic uses

- See the following slide.

Actions → adverse effects and cautions

- Adverse effects are predictable because of well-understood mechanisms of action and widespread distribution of adrenergic receptors. They mimic the actions of sympathetic nervous system stimulation leading to a wide range responses.
- Intensity of effects is dose-dependent. Higher doses lead to exaggerated responses.
- When administered together, sympathomimetics enhance the effects of other drugs with sympathomimetic effects potentially leading to hypertensive crisis.

Key points: Therapeutic uses

- Sympathomimetics are an important group of drugs used for cardiovascular, respiratory, and other conditions. They are classified into subgroups based on their spectrum of action: directing action by binding and activating adrenergic receptors (alpha, beta, and/or dopamine), indirect action by increasing synaptic levels of endogenous norepinephrine, which activates adrenergic receptors, or mixed-actions (both direct and indirect actions).
- Endogenous adrenergic receptor agonists epinephrine, norepinephrine, and dopamine are available as pharmacotherapeutics – vasopressors (induce vasoconstriction).
- Epinephrine is the drug of choice in acute anaphylaxis. It prevents or reverses airflow obstruction due to bronchoconstriction, prevents or reverses cardiovascular collapse, and decreases release of mediators from mast cells.
- Norepinephrine is the first-line agent for the initial treatment of shock (dangerously low hypotension and inadequate organ perfusion – life-threatening).
- Dopamine is an alternative for norepinephrine in selected patients with shock.
- Phenylephrine is a selective alpha-1 receptor agonist for treatment of patients with shock that have failed therapy with other vasopressors. It is also used topically for treatment of nasal and conjunctival congestion.
- Isoproterenol is rarely used.
- Dobutamine, a beta-1 receptor agonist, is administered for short-term use in decompensated heart failure for its inotropic effects.

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Must-know terms and mechanisms to understand adrenergic pharmacology

Catecholamines	Neurotransmitters derived from the amino acid tyrosine: dopamine, norepinephrine (NE), and epinephrine (EPI)
Neurotransmitters of the SNS	NE and EPI (a hormone) are synthesized from dopamine. • NE is synthesized in presynaptic neurons of the SNS, and released from SNS terminal boutons when the SNS is activated. • EPI is a hormone synthesized from NE in chromaffin cells of the adrenal medulla and released into the bloodstream when the SNS is activated. EPI travels through the blood stream to sites of action. Dopamine in the periphery is released from cells of the kidney and mesenteric tissues (especially GI tract).
Adrenergic	Nerve cells or fibers that release or are activated by norepinephrine (noradrenalin) and/or epinephrine (adrenalin)
Adrenergic receptors	α1ARs, α2ARs, β1ARs, β2ARs, β3ARs
Termination of neurotransmitter effect	• NE is taken back up into the presynaptic neuron via NE transporters (NET) and re-stored in vesicles or degraded by monoamine oxidase (MAO). • EPI is degraded primarily in the liver and other tissues by MAO and catechol-O-methyltransferase (COMT).

Pharmacological targets of adrenergic drugs

α1AR	GqPCRs: ↑ IP3 and DAG: Smooth muscle contraction Blood vessels, iris radial muscle, pilomotor, urinary tract, prostate, pregnant uterus
α2AR	GiPCRs: Inhibit adenylyl cyclase and ↓ cAMP: Inhibitory actions Widely distributed in periphery and CNS: presynaptic auto- and heteroreceptors; Pre- and post-synaptic neurons in CNS; vascular smooth muscle (cause vasoconstriction)
β1AR	GsPCRs: Activate adenylyl cyclase and ↑ cAMP: Activation by NE and EPI Present on postsynaptic membranes; predominate in the heart → ↑ heart rate, conduction velocity, and force of contraction
β2AR	GsPCRs: Activate adenylyl cyclase and ↑ cAMP: Activation by EPI Widely distributed on extra-junctional sites on vascular smooth muscle in skeletal muscle beds (dilation), bronchial smooth muscle (dilation), skeletal muscle (itself), pancreas, platelets, immune cells; pre- and postsynaptic CNS neurons Extra-junctional sites: non-synaptic sites
β3AR	GsPCRs: Activate adenylyl cyclase and ↑ cAMP: Activation by NE and EPI Present on postsynaptic membranes; expressed by adipocytes (thermogenesis); detrusor muscle of urinary bladder (relaxation); cardiomyocytes (function unclear)

Sympathetic responses

chronotropy inotropy

Heart:
Cardiac excitatory action induces ↑ heart rate and ↑ force of contraction; mediated by β1ARs

Vasculature:
EPI-induced vasodilation of smooth muscle of vasculature in skeletal muscle, mediated by β2ARs. Constriction of vascular smooth muscle of vessels supplying skin and splanchnic region → shunts blood flow from these areas to skeletal muscles.

Metabolism: ↑ rate of hepatic glycogenolysis and gluconeogenesis and liberation of free fatty acids from adipose tissue → blood glucose rises; mediated by β2ARs.

Bronchiolar smooth muscle: Relaxation → bronchodilation; mediated by β2ARs

Eyes: Pupils dilate → mydriasis; mediated by α1ARs
Genitourinary tract: Relaxation of detrusor muscle of bladder (β3ARs, promotes urine storage); maintain tonic contraction of internal urethral sphincter (remains closed) (α1ARs); contraction of prostatic smooth muscle, which increases urethral resistance (α1ARs)

CNS: Signals received in higher brain centers facilitate purposeful responses or imprint events in memory.

Cardiovascular terminology in this presentation

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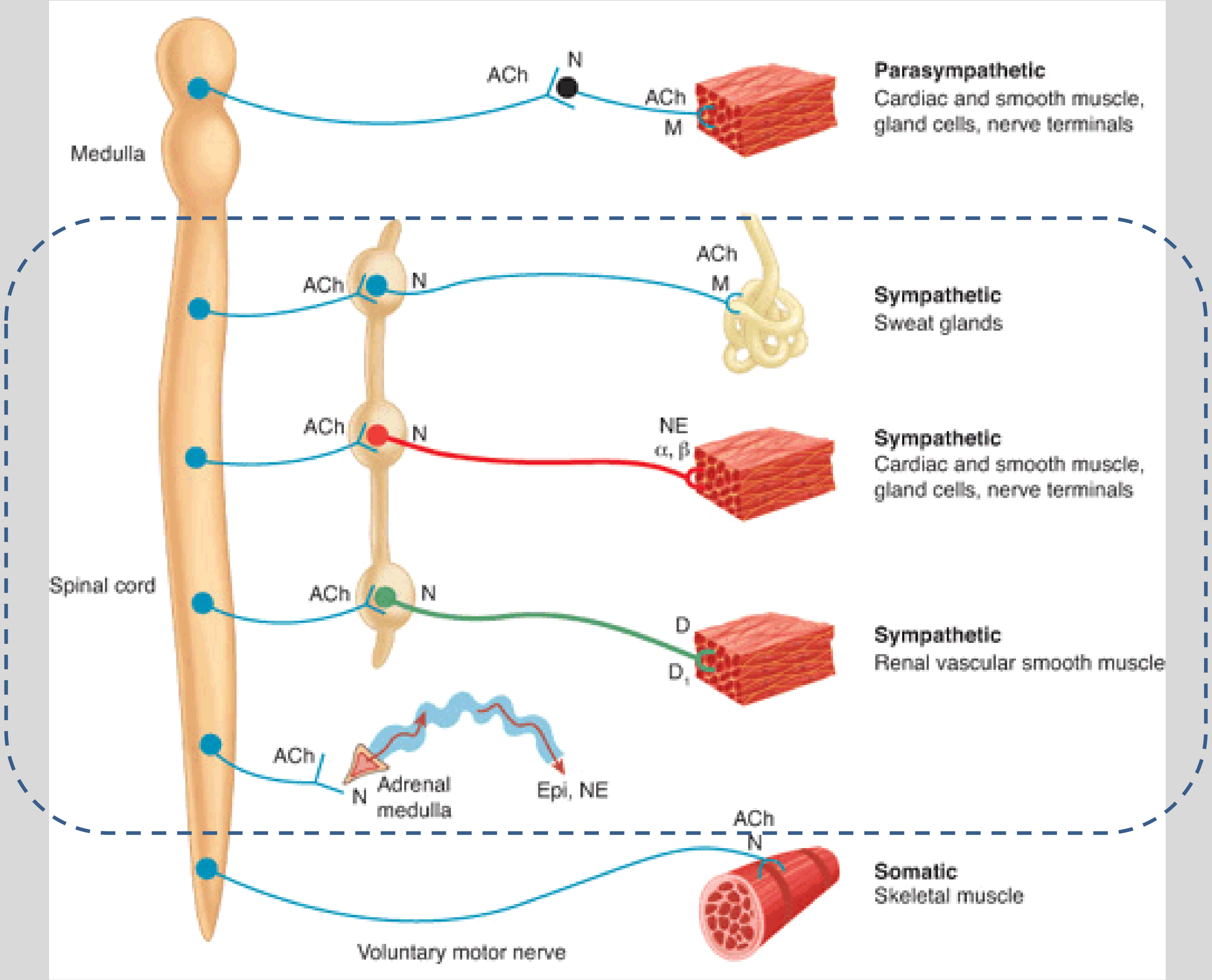
Heart rate (HR)	The number of times the heart beats per minute. Normal HR is 60-100 bpm.
Stroke volume (SV)	The amount of blood pumped out of the left ventricle in one heart beat.
Cardiac output (CO)	The total amount of blood the heart pumps per minute: HR x SV = CO
Peripheral resistance Total peripheral resistance (TPR)	The resistance by blood vessels against the flow of blood. Total peripheral resistance is the amount of force the entire systemic vasculature (except the pulmonary circulation) exerts on blood flow. The main contributors are the smallest arteries and arterioles.
Mean arterial pressure (MAP)	The average arterial pressure throughout one cardiac cycle – systole and diastole – is influenced by cardiac output and total peripheral resistance.
Preload	The volume of blood in the ventricles just before they contract. Preload is influenced by the venous return to the heart and the compliance of the ventricles.
Afterload	The pressure (the load) the heart must overcome to push blood out of the ventricles into the systemic circulation. Afterload is primarily determined by the total peripheral resistance and the aortic pressure.

Cardiovascular terminology in this presentation

Systolic blood pressure (SBP)	BP increases to maximum pressure as left ventricular ejection (systole) of blood into aorta increases pressure and distends the vascular wall, which creates a tension wave along the artery wall. SBP is influenced by cardiomyocyte contraction, stroke volume, arterial compliance, and total peripheral resistance.
Diastolic blood pressure (DBP)	The minimum pressure in the arteries between heartbeats when ventricles relax and refill with blood (diastole) is influenced by total peripheral resistance, blood volume, and blood viscosity.
Pulse pressure	The difference between SBP and DBP (SBP – DBP = pulse pressure) (typically 40 to 60 mmHg in a young healthy adult)
Baroreceptors	Receptors located in the carotid sinuses and aortic arch are sensitive to changes in the stretch of arterial walls that occur with changes in mean arterial pressure to maintain blood pressure within normal levels.

Cardiovascular terminology in this presentation

Reflex responses to changes in BP	• When blood pressure rises, baroreceptors detect greater stretch in the arterial walls and send a signal via glossopharyngeal nerve (CN IX) to the brainstem, which increases vagal signaling (CN X), resulting in a lower heart rate (bradycardia) and vasodilation, which lowers blood pressure. • When the blood pressure decreases, baroreceptors send a signal to the brainstem, which increases sympathetic signaling resulting in increased heart rate (tachycardia) and vasoconstriction, raising blood pressure.
Bradycardia	Heart rate <60 bpm
Tachycardia	Heart rate >100 bpm (sinus tachycardia)
Renin (ree′ nin)	An enzyme secreted from specialized cells in the kidney when blood pressure drops too low. Renin increases levels of angiotensin II, a powerful vasoconstrictor and stimulator of aldosterone secretion. Aldosterone causes reabsorption of sodium and water (RAAS – renin-angiotensin-aldosterone system). Increased sympathetic tone activates β1ARs in the kidneys, which induces renin release, activating the RAAS.

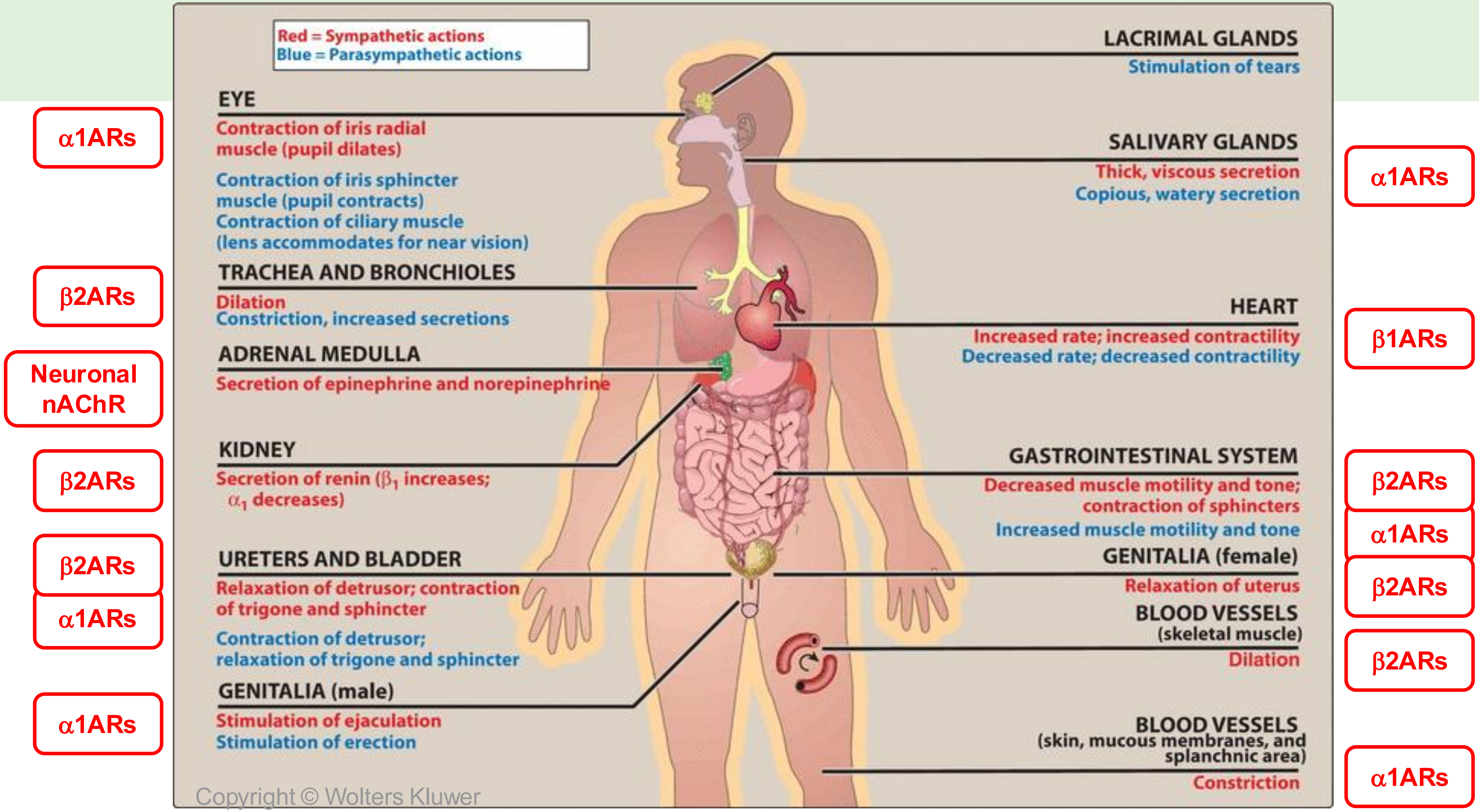


Katzung & Vanderah's Figure 6-1

- The sympathetic nervous system controls cardiovascular effects and activities that increase arousal and energy expenditures and inhibit digestion.

LWW Health Library

From: **3 The Autonomic Nervous System** Lippincott® Illustrated Reviews: Pharmacology, 7e, 2019



Actions of sympathetic and parasympathetic nervous systems on effector organs.

The pattern: Stimulation of α1ARs on smooth muscle causes contraction.

Stimulation of β2ARs on smooth muscle causes relaxation.

Neuronal regulation of the heart

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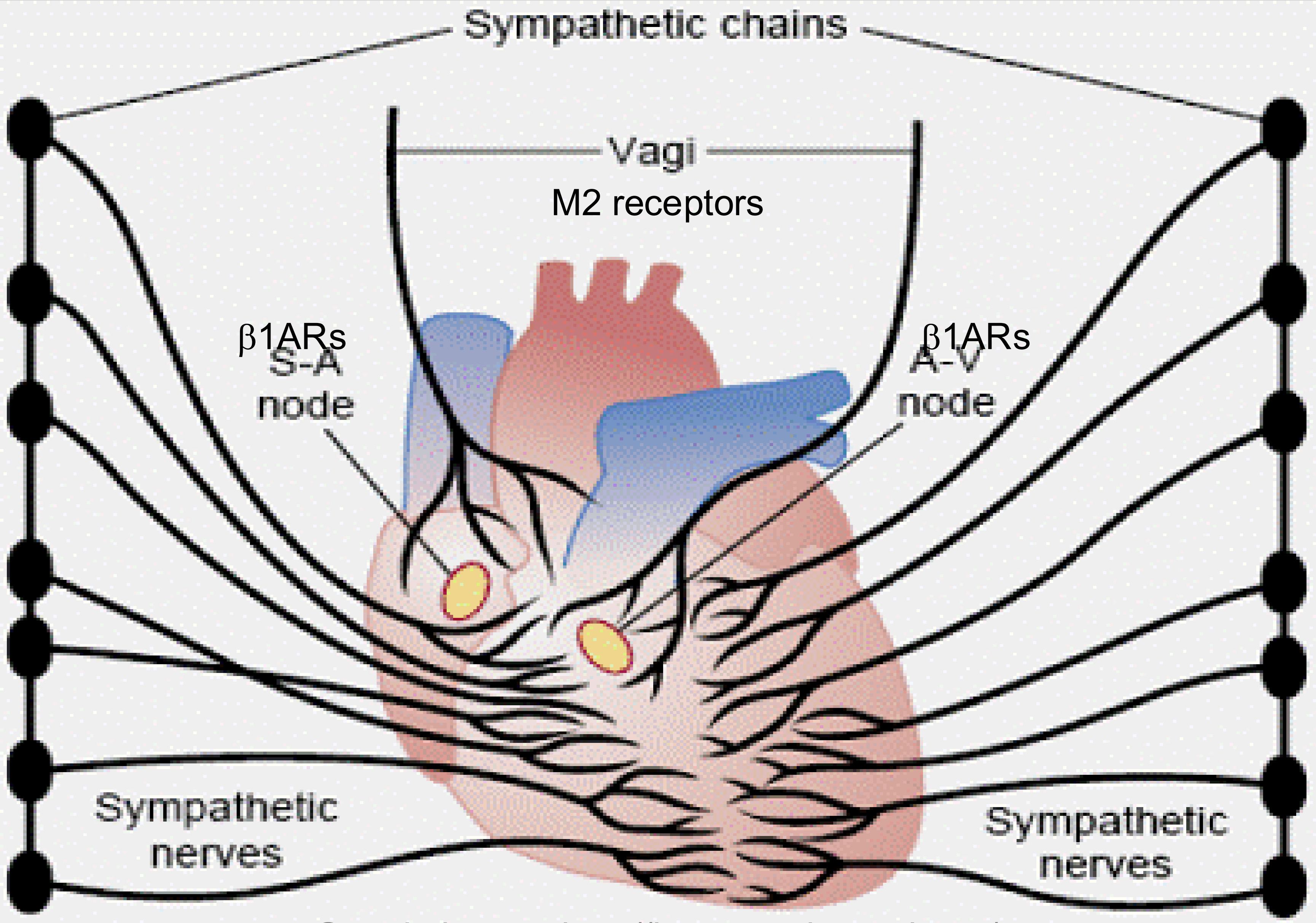
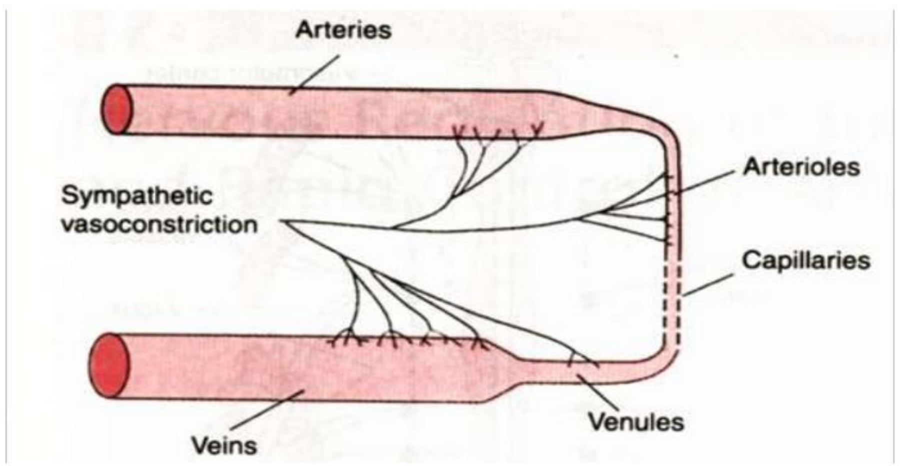


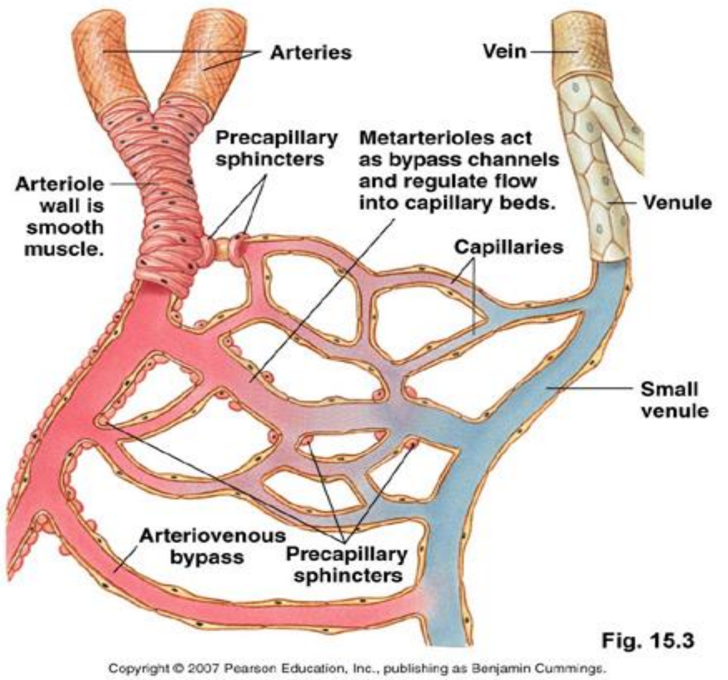
Google image: http://intranet.tdmu.edu.ua/

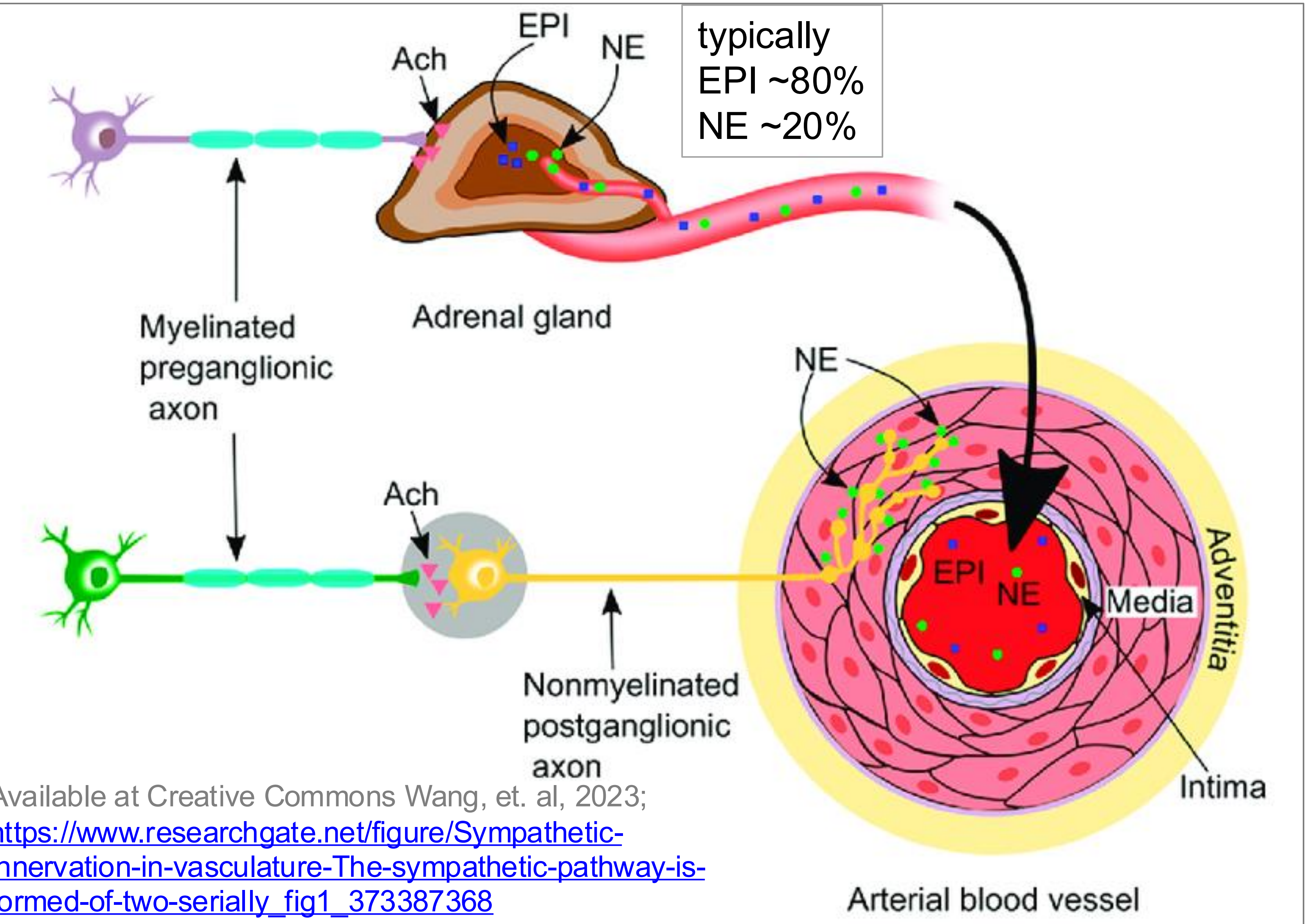
SA node: Stimulation of **β1 adrenergic receptors** in the sinoatrial node expresses increases heart rate (positive chronotropy).
AV node: Stimulation of **β1 adrenergic receptors** in the atrioventricular node speeds conduction (positive dromotropy).
Stimulation of **β1 adrenergic receptors** in the myocardium enhances contractility (positive dromotropy).
Stimulation of **β1 adrenergic receptors** in the myocardium also improves relaxation (positive lusitropy).

Autonomic innervation of the blood vessels



http://www.slideshare.net/ananthatiger/blood-pressure-mechanism





Available at Creative Commons Wang, et. al, 2023; https://www.researchgate.net/figure/Sympathetic-innervation-in-vasculature-The-sympathetic-pathway-is-formed-of-two-serially_fig1_373387368

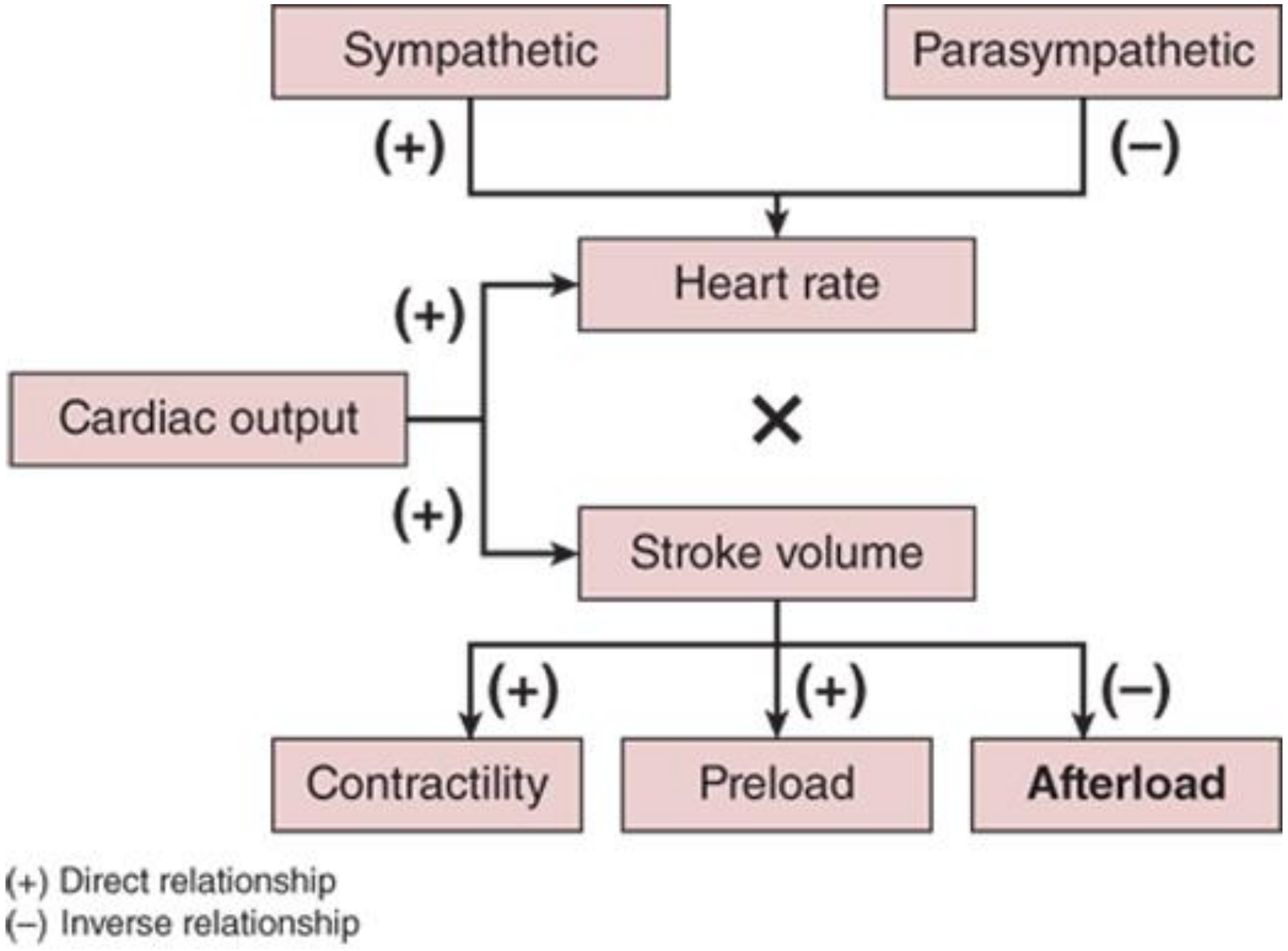
Stimulation of α**1ARs** on vascular smooth muscle induces vasoconstriction.
Stimulation of post-synaptic α2ARs also induces vasoconstriction, but these are less abundant in the periphery than α1ARs.
Stimulation of β2ARs on skeletal muscle vascular smooth muscle induces vasodilation.

Main factors regulating cardiac output.

- **Cardiac output** (CO) varies based on the body's requirements and depends on the heart rate (beats per minute) and the stroke volume (blood volume pumped per heartbeat):

$$CO = HR \times SV$$

- **Heart rate** (HR) is determined by the balance between sympathetic and parasympathetic activation, normally, on the sinoatrial (SA) node of the heart.
- **Stroke volume** (SV) is the volume of blood ejected from each ventricle during the contraction in 1 heartbeat. It is equivalent to the difference between the volume of blood in the ventricle just before the contraction (end-diastolic volume) and the volume of blood left in the ventricle after the contraction (end-systolic volume).
- **Preload** is defined as the stretching of the myocardial muscle fibers just prior to a contraction or ventricular wall tension at the end of diastole. The more blood returning to the heart (within physiologic limits) as venous return, the more blood pumped from the heart as cardiac output.

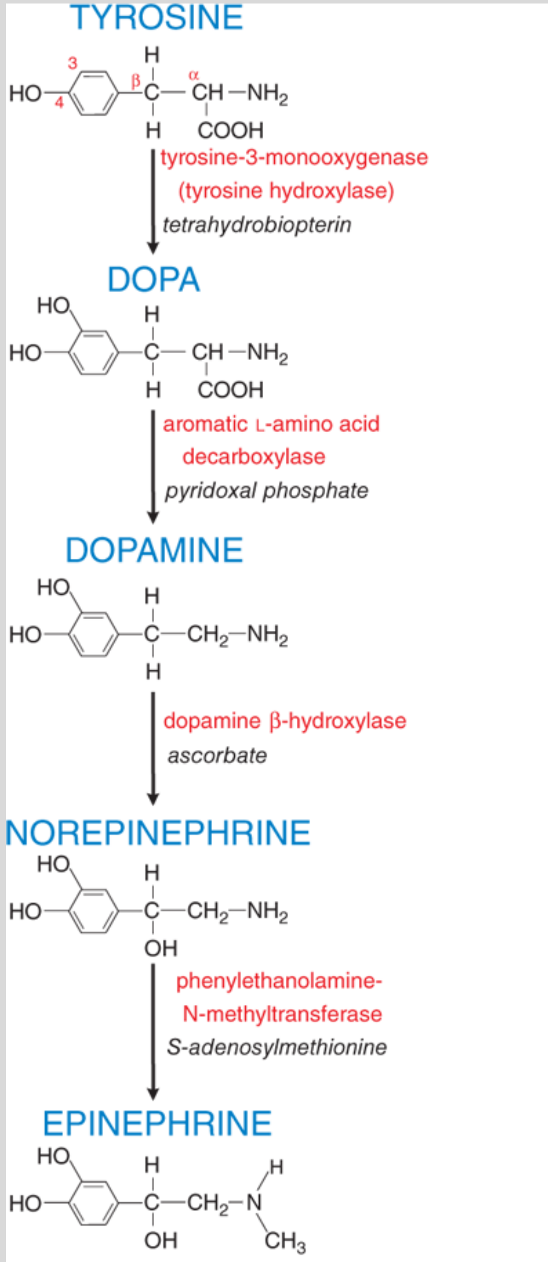


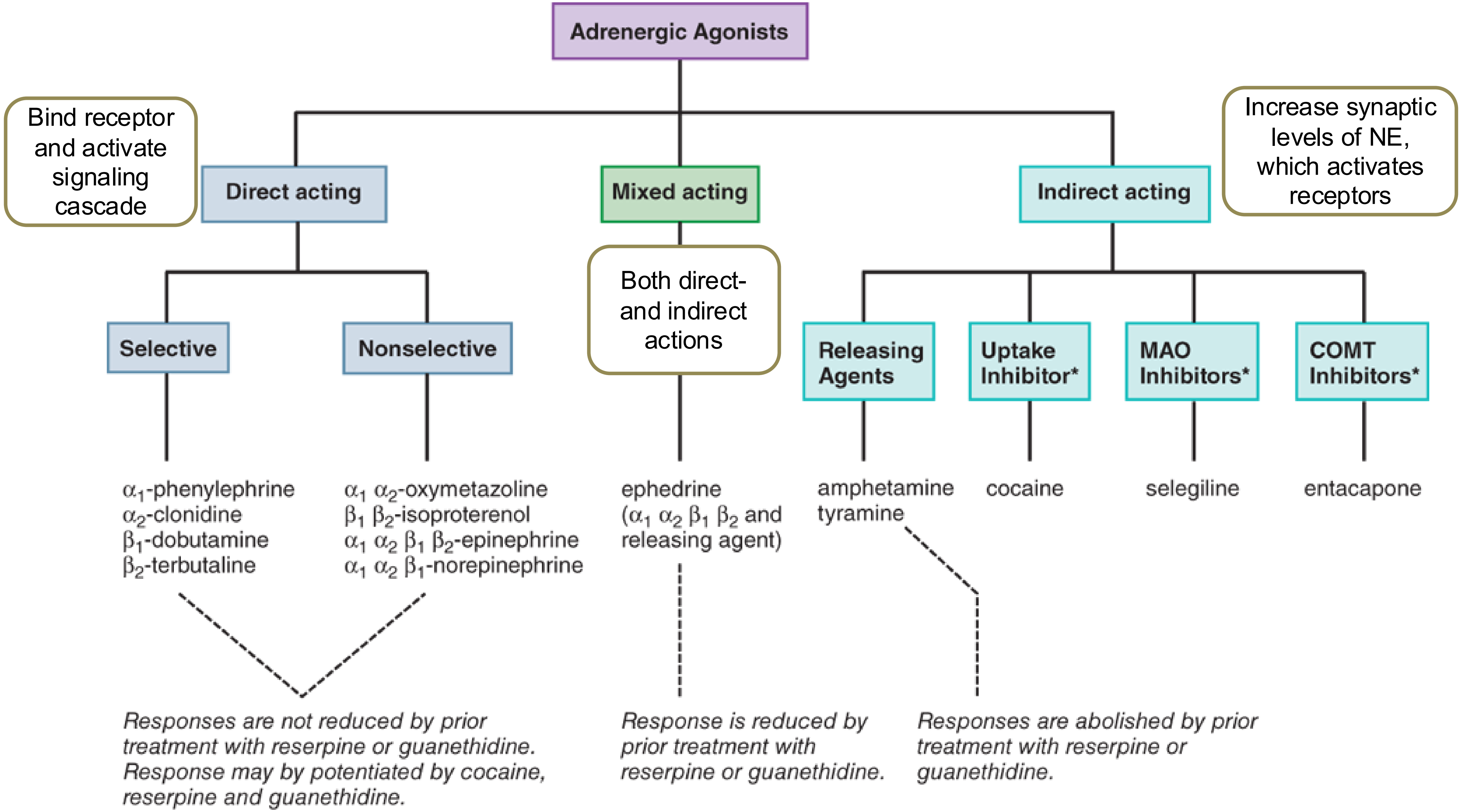
Source: Adel Elmoselhi:
Cardiology: An Integrated Approach

- **Afterload** is defined as the ventricular wall stress during the ejection phase of systole or the resistance that the ventricle needs to overcome in order to eject its content – the amount of aortic pressure that the heart ejects blood against in order to empty the left ventricle. Afterload is determined by several factors, including wall stress, aortic pressure, and total peripheral resistance.

Citation: Chapter 5 Cardiac Output and the Pressure-Volume Relationship, Elmoselhi A. *Cardiology: An Integrated Approach;* 2017. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=171660171&bookid=2224&Resultclick=2 Accessed: November 11, 2024

Pharmacology of Sympathomimetics





Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition:

Classification of adrenergic receptor agonists (sympathomimetic amines) or drugs that produce sympathomimetic effects. For each category, a prototypical drug is shown.

Citation: Chapter 12 Adrenergic Agonists and Antagonists, Brunton LL, Hilal-Dandan R, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e;* 2017, figure 12-1. Available at: https://accessmedicine.mhmedical.com/ViewLarge.aspx?figid=194543661&gbosContainerID=0&gbosid=0&groupID=0&sectionId=167890123&multimediaId=undefined

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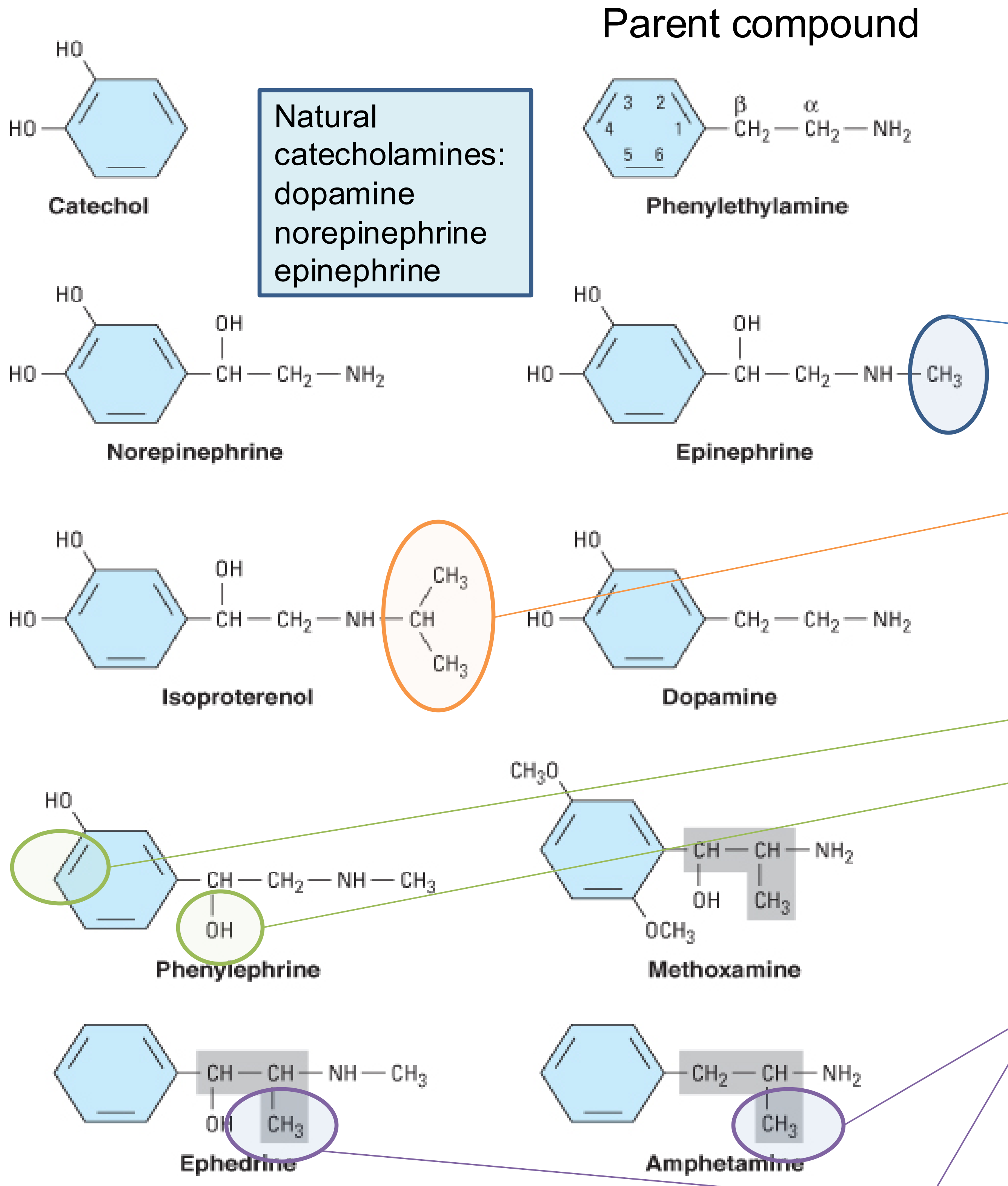
Structure-Activity Relationships



Katzung Fig 9-4, 9-5

Epinephrine: methyl substitution on amino group enhances β_2 activity

Isoproterenol has larger substitution on the amino group → enhanced βAR activity + very weak activity at αARs.

Phenylephrine
- No C4 –OH on benzene ring: βARs low affinity
- Resistance to metabolism by COMT
- Presence of C3 –OH: α1AR selectivity

Ephedrine and Amphetamine
- Substitution on α carbon → ↑lipophilicity, ↑oral bioavailability, ↑CNS activity
- Methyl substitution on α carbon →↑resistance to MAO (↑ duration of action) + displaces NE from storage sites

Direct-Acting Sympathomimetics

Vasopressors	Inotropes	Bronchodilators	Bladder relaxants	Alpha-2 agonists
✳Epinephrine	✳Isoproterenol	✳Albuterol	✳Mirabegron	✳Clonidine
✳Norepinephrine	✳Dobutamine	Levalbuterol	Vibegron	✳Methyldopa
✳Dopamine		✳Salmeterol		
✳Phenylephrine		Several others		

Vasopressors

Powerful drugs that raise the mean arterial pressure (MAP) by inducing vasoconstriction.

Vasopressors are used in the treatment of shock, a life-threatening condition of circulatory failure and a medical emergency.

- Epinephrine
- Norepinephrine
- Dopamine
- Phenylephrine

The drugs are differentiated by their relative affinities for the adrenergic receptors.

✳Epinephrine

A potent stimulant of both α and β adrenergic receptors

Relative affinities	$\beta_1 = \beta_2 > \alpha_1 = \alpha_2$

Endogenous Epinephrine's Physiologic Effects ("Fight or Flight" response)

Summarizing EPI's physiologic actions and effects provides context for pharmacologic EPI's clinical activity.

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Cardiac (β1ARs)	• ↑ heart rate • ↑ stroke volume • ↑ cardiac output • ↑ coronary blood flow • proarrhythmic	Respiratory (β2ARs)	• Bronchodilation
Blood pressure (cardiac &vascular)	• ↑ systolic blood pressure • usually ↓ diastolic pressure • MAP only slight change • ↑ mean pulmonary pressure	Metabolic	• ↑ oxygen consumption • ↑ blood glucose • ↑ lactic acid • ↓ plasma K^+
Peripheral circulation (α1&β2 ARs)	• ↓ or unchanged TPR • slight ↑ cerebral blood flow • ↑skeletal muscle blood flow (β2) • ↓ cutaneous blood flow (α1) • ↓ splanchnic blood flow (α1)	CNS	• ↑ respiration • ↑ subjective sensations
		Immune system	• Various immunomodulatory actions

Epinephrine's Cardiovascular Effects

Dose-dependent effects in the vasculature

- Epinephrine in **low doses** acts preferentially on **β_2 receptors** located in the skeletal muscle vascular beds → **vasodilation in skeletal muscle vascular beds** → ↓ total peripheral resistance → ↓ diastolic blood pressure
  - **α-Mediated** vasoconstriction in the splanchnic / kidney blood vessels and skin vessels shunts blood to the skeletal muscle vasculature.
- **High-dose** EPI causes **vasoconstrictive α1 effects** in arteries and veins, including in the skeletal muscle vasculature, **to dominate** → profound increase in mean arterial pressure (MAP).

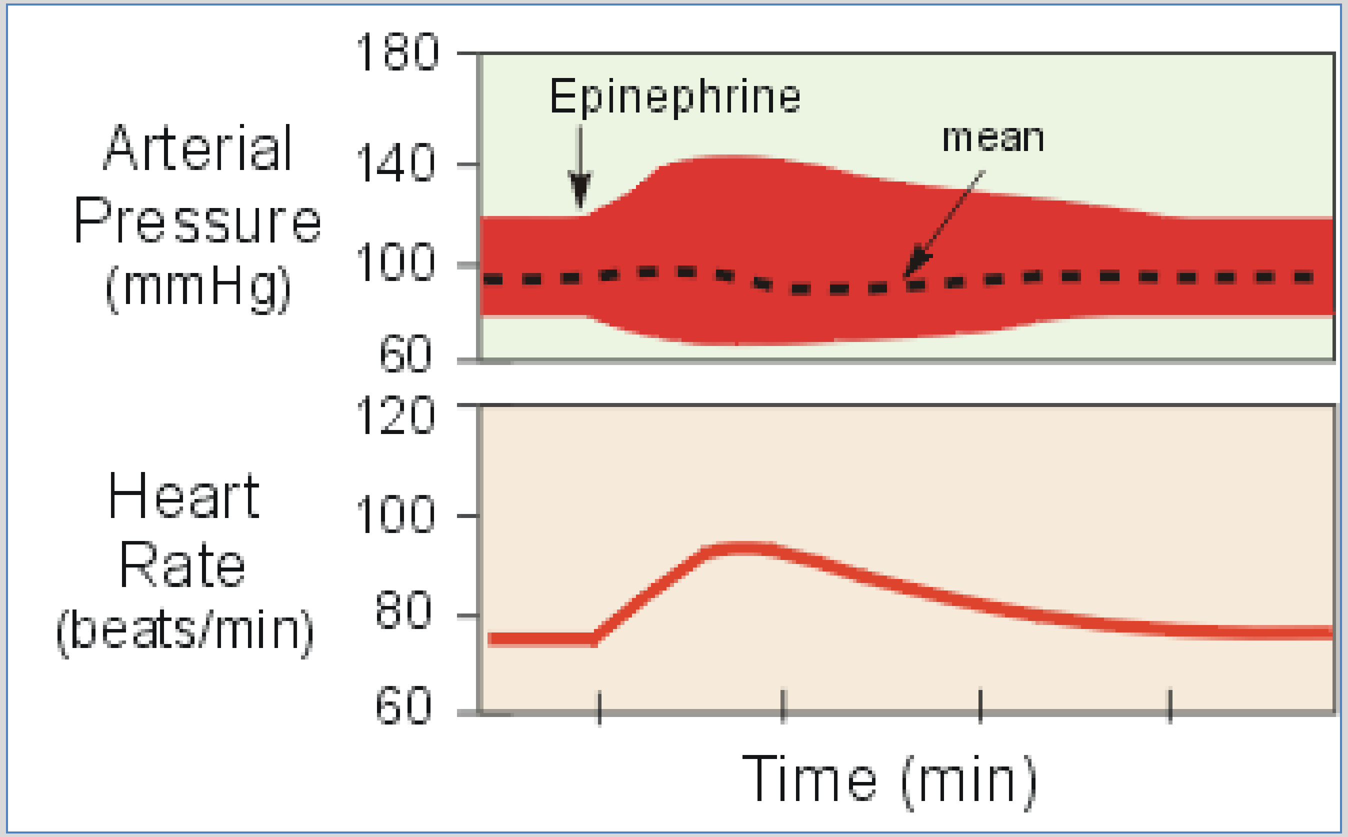
Direct myocardial stimulation →

- ↑ heart rate (positive chronotropic action)
- ↑ conduction velocity
- ↑ strength of ventricular contraction (positive inotropic action)
- ↑ ventricular work per beat is increased → ↑oxygen → risk of ischemia in patients with coronary artery disease (damage from plaques)

Epinephrine's dose-dependent effects on blood pressure

1. Low dose epinephrine

- **β_2-mediated** enhanced blood flow to skeletal muscles → ↓peripheral resistance → ↓diastolic blood pressure
- **β_1-mediated** ↑heart rate (HR) and ↑force of contraction → ↑stroke volume (SV), ↑cardiac output, ↑cardiac oxygen demand
- **Mean blood pressure** (MAP) usually not greatly elevated with low-dose EPI



http://www.cvphysiology.com/Blood%20Pressure/BP018.htm

Little change in MAP → compensatory baroreceptor reflexes do not appreciably antagonize direct cardiac actions.

EPI effects on BP at intermediate and rapid administration infusion rates

2. Intermediate rate of epinephrine infusion

↓

greater degree of
α-mediated vasoconstriction

↓

compensatory reflexes
may be activated as MAP rises

3. Rapid infusion of epinephrine
i.e. high concentration (DANGEROUS)

↓

predominantly α effects
(including skeletal muscle vasculature)

↓

↑ TPR + ↑HR and ↑contractility
α-mediated vasoconstriction; β-mediated cardiac effects

↓

↑ systolic and diastolic blood pressure

↓

Reflex homeostatic adjustments
→ ↓ heart rate

Epinephrine Pharmacokinetics

A	Epipen IM autoinjector, subQ, IV, IO, intranasal spray, oral inhalation • **IV / IO route: Caution**; Careful use by IV or IO route *only* if immediate, reliable effect is required (*eg* profound hypotension or cardiopulmonary arrest) • Not absorbed orally – rapidly metabolized in GI mucosa and liver IO: intraosseous
D	Catecholamines do not cross the blood-brain barrier.
M	Circulating epinephrine: MAO, COMT degradation in many tissues, particularly the liver
E	Metabolites and small amount of unchanged compound normally excreted in urine • *Clinical correlate:* High levels of Epi, NE, and metabolites in blood and urine are indicative of pheochromocytoma.
onset	IM rapid; subQ ~5-10 minutes; intranasal spray ~1 minute IV immediate

Epinephrine Therapeutic Uses and Class Adverse Effects

Therapeutic uses:

- IM and intranasal spray (without delay!)
  - **Anaphylactic shock reversal**

Drug of Choice

never contraindicated

for anaphylaxis treatment

- Subcutaneously with local anesthetics
  - local vasoconstriction
- Oral inhalation via nebulizer
  - acute asthma bronchodilation
- IV for cardiovascular effects:
  - myocardial stimulation during cardiopulmonary resuscitation
  - alternative to NE in septic shock
- Topical hemostat

Adverse effects:

- Restlessness, apprehension, throbbing headache, tremor, palpitations
- Premature ventricular contractions
- Ventricular arrhythmias due to excessive stimulation
- Angina (↑O_2 consumption)
- Cerebral hemorrhage
  (large dose → sharp ↑ BP)

Cautious use:

Conditions when response to EPI could lead to severe hypertension or cardiac complications

Adverse effects:

Sympathetic stimulation

Extravasation at infusion site (leakage) → tissue necrosis

Check your knowledge

Where is EPI synthesized and secreted from (in the periphery)?

What are EPI's relative affinities for adrenergic receptor types?

What are the effects of physiologic EPI on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

Check your knowledge

EPI is the drug of choice for what condition?

Why is it the drug of choice (actions and effects)?

✳Norepinephrine

A potent stimulant of α and β1 adrenergic receptors

Relative affinities for ARs			
NE	Epi compared to NE		
$\alpha_1 = \alpha_2 > \beta_1$	α: Epi > NE	β_1: Epi = NE	β_2: Epi >>>>>NE

Norepinephrine Effects and Therapeutic Uses

Potent vasopressor (α effects)

- Vasoconstriction increases total peripheral resistance → ↑afterload
- Constriction of veins → ↑venous return to the heart → ↑preload and stroke volume

Cardiac effects (β1 effects)

- ↑cardiac force of contraction → modest ↑cardiac output
- ↑MAP → reflex parasympathetic effect → heart rate unchanged or decreased

THERAPEUTIC USES: Circulatory shock

Profound hypotension with low peripheral resistance

- Septic | Cardiogenic | Neurogenic

drug of choice for initial therapy

based on clinical experience of effectiveness

- ➢ *Selection of therapy depends on patient-specific factors.*
- ➢ *Provide adequate blood volume restoration.*

ADVERSE EFFECTS typical of the class:

- Reduced blood flow to organs → ↓ organ perfusion and danger of organ failure
- Ventricular arrhythmia
- Digital ischemia → gangrene
- Extravasation at injection site → tissue necrosis, gangrene

Check your knowledge

Where is NE synthesized and secreted from?

What are NE's relative affinities for adrenergic receptors?

What are the effects of intravenous NE on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

Check your knowledge

NE is the drug of choice for what conditions?

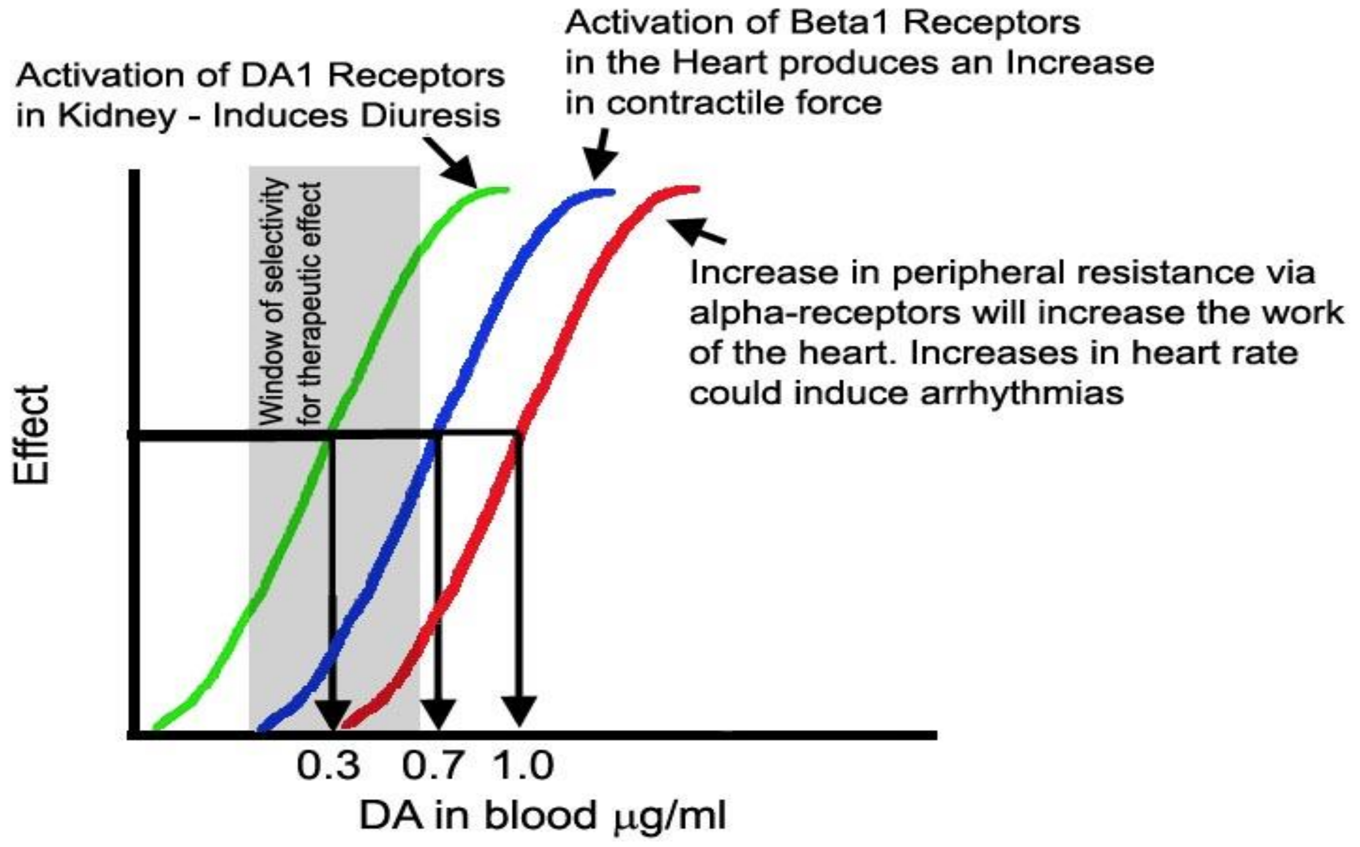
Why is it the drug of choice (actions and effects)?

- Dopamine

Stimulates peripheral D1, α1 and β1 receptors

Relative affinities for ARs			
DA	α_1	β_1	β_2
$D_1 > \beta_1 > \alpha_1$	EPI > NE >> DA	EPI > NE = DA	EPI >>>> DA

Dopamine's Dose-Dependent Hemodynamic Effects



Low dose (1-5 mcg/kg/min)

D_1 activation in renal and mesenteric vessels → ***vasodilation***

- ↑ renal blood flow, ↑ GFR, and ↑ Na^+ excretion
- ↓ peripheral vascular resistance

Intermediate dose (5-10 mcg/kg/min)

β_1 ARs activation → ↑ contractility (direct effect) + causes NE release from nerve terminals (indirect effect) → positive inotropic effect

- ↑ systolic blood pressure; diastolic pressure same or slightly increased
- Total peripheral resistance usually unchanged

High-dose (>10 mcg/kg/min)

α_1 ARs activation → general vascular constriction (pressor effect)

Therapeutic use: alternative to NE for septic and cardiogenic shock

- Dopamine was associated with higher mortality and dysrhythmias were more common (subgroup analysis from a randomized trial)
- Dopamine does not appear to provide any renal protecitve advantage over routine therapy with IV diuretics.

✳Phenylephrine: α1-selective vasopressor not a catecholamine – not inactivated by COMT

PK properties

A	Oral, IV, IM, subQ, intranasal, intraocular (systemic absorption can occur)
M	Hepatic and gut wall: MAO, sulfation, glucuronidation Bioavailability, oral ≤38% bioavailability
t½	I.V. ~15 min; I.M. or subQ 1-2 hours ophthalmic time to recovery 3-8 hours

Phenylephrine Therapeutic Uses

Hemodynamic Effects

Potent vasoconstriction

↓

- ↑ peripheral resistance
- ↑ SBP and ↑DBP
- Marked reflex bradycardia
- May decrease SV

Therapeutic uses	Route	Comments
Vasopressor	I.V.	treatment of hypotension (alternative in patients who cannot take NE or failed other therapies)
Nasal decongestant ➢ Oral phenylephrine in OTC cough and cold products is ineffective.	Intranasal	vasoconstriction of venous capacitance vessels in nasal tissues *Intranasal → Rebound congestion*
Ocular decongestant	topical	relief of “red eyes”
Mydriatic	topical	activation of pupillary sphincter
Hemostatic agent	topical	local vasoconstriction
Hemorrhoids	topical	local vasoconstriction

BP: blood pressure
SBP: systolic BP
DBP: diastolic BP
SV: stroke volume

Check your knowledge

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What are the dose-dependent effects of intravenous dopamine infusion on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

What are the effects of intravenous phenylephrine infusion on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

What are the clinical indications for dopamine and phenylephrine?

Positive Inotropes

Sympathomimetics that increase the force of cardiomyocyte contraction and are used primarily for their positive inotropic effects.

- Isoproterenol
- Dobutamine

Isoproterenol

Relative affinities	**Potent, nonselective βAR agonist**
activity at β_1AR, β_2AR and β_3AR	No affinity for αARs
β_1: ISO >> EPI > NE = DA	β_2: ISO > EPI >>> NE > DA

- Cardiac: positive **chronotropic** and **inotropic** actions → ↑cardiac output and ↑O_2 consumption
- Nonvascular: β_2-mediated relaxation of bronchial and GI smooth muscle

β_2AR stimulation →

- ↓peripheral vascular resistance
- ↓diastolic BP
- systolic BP may be unchanged or rise
- ↓ MAP

Limited Therapeutic Uses: Treatment of hypotension resulting from bradycardia

Adverse Effects

- Reflex tachycardia → Prominent chronotropic effect

Cardiac ischemia / arrhythmias can occur in patients with underlying coronary or cardiac disease.

Dobutamine: “Selective” $\beta 1$- agonist

(a semisynthetic catecholamine metabolized by liver and tissue COMT)

Racemic mixture of both entantiomeric forms of the drug:

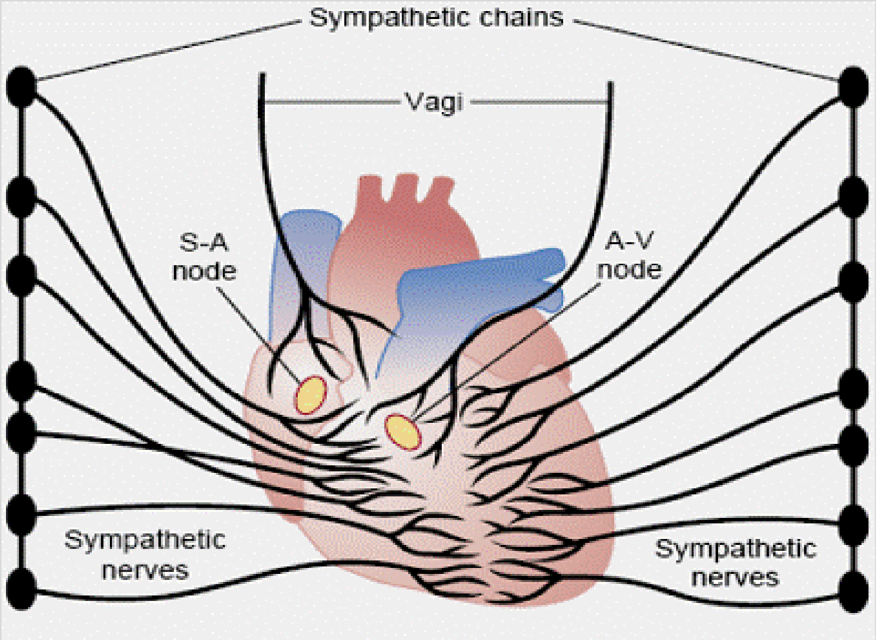
Potency of (+) isomer on βARs ~10x > than (-) isomer

The αAR effects offset each other so the β_1AR effects predominate:

1. The (-) isomer is a potent α_1AR agonist → marked vasoconstrictor effects.
2. The (+) isomer is an α_1AR antagonist → may offset the agonist effects of (-) isomer.
3. The (+) isomer is a potent β_1AR agonist.
4. The (+) isomer also has mild β_2AR agonist activity.

Dobutamine: β Agonist **Net Effects**

- Dobutamine has predominantly β_1 receptor agonist action and a mild β_2 receptor agonism.
- It causes β_1-mediated increased cardiac contractility and heart rate, resulting in increased cardiac output (CO = HR x stroke volume).
- Its β_2 activating effects may be slightly greater than its α_1 stimulating effects resulting in some vasodilation.



An advantage of dobutamine use compared to isoproterenol is increased cardiac output usually with a relatively mild increase in heart rate due to less reflex tachycardia.

Summary of dobutamine's effects

↑ cardiac output	↓ systemic vascular resistance (↑CO → reflex vasodilation + $\beta 2$ effect)
↓ central venous pressure and ↓ pulmonary wedge pressure (indicators of preload)	improves renal blood flow

At low doses, dobutamine relieves the signs and symptoms of congestive heart failure.

THERAPEUTIC USES

- Cardiac decompensation,
  due to cardiac surgery, CHF, acute MI
  - short-term treatment
- Dobutamine stress testing in patients unable to perform exercise stress testing (with echocardiogram)

➢ Long-term dobutamine use is associated with decreased survival.

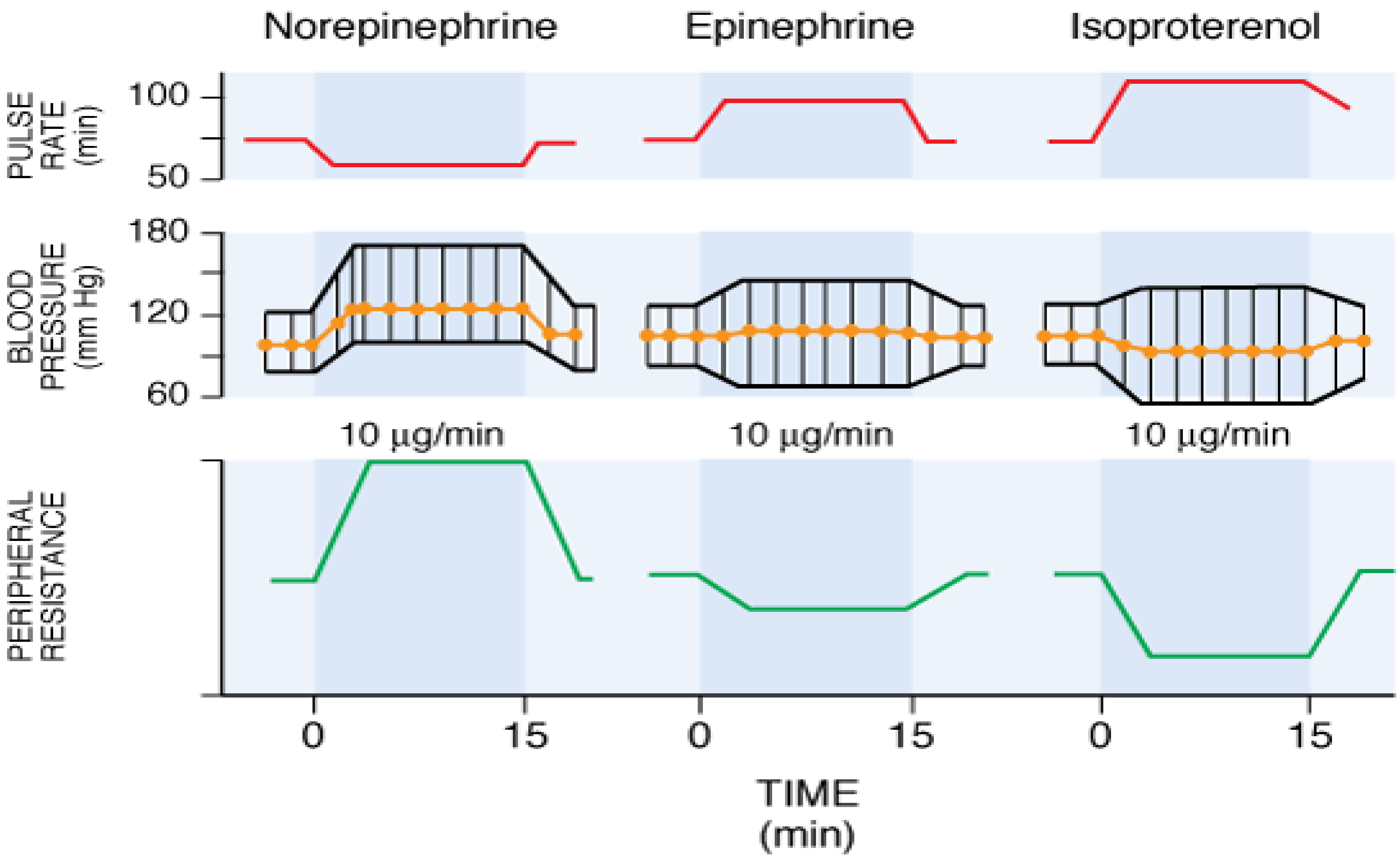
CHF: congestive heart failure; MI: myocardial infarction

ADVERSE EFFECTS

- Tachyphylaxis:
  24h-72h continuous use → agonist-induced βAR desensitization / receptor downregulation requiring dose adjustment
- Dose-dependent rise in systolic pressure and heart rate → hypertension, tachycardia, angina, arrhythmia
  Hypertension, angina, arrhythmia, tachycardia related to the β_1 actions

Sympathomimetics

- Test your knowledge by comparing the cardiovascular effects of norepinephrine, epinephrine, and isoproterenol.



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*: www.accessmedicine.com

Comparing the cardiovascular effects of norepinephrine, epinephrine, and isoproterenol

Comparing the effects of NE, EPI, and isoproterenol

NE: α–mediated vasoconstriction → ↑TPR + ↑SBP and ↑DBP and ↑MAP

- Potent vasoconstriction causes an increase in MAP → which causes a modest increase in heart rate (direct activation of β_1 in the SA node) → followed by reflex bradycardia in response to increased MAP → which results in a heart rate that is unchanged or mildly decreased.

EPI (physiologic concentrations):

β_1 activation → ↑ rate and force of cardiac contraction → ↑SBP

- β_2-mediated vasodilation in skeletal muscle vascular beds → ↓TPR → ↓DBP
- MAP is modestly increased due to α-mediated vasoconstriction in other vascular beds so is *unlikely to trigger the reflex effect* on heart rate.
- Activation of β_1 cardiac receptors → ↑HR and ↑inotropy

Isoproterenol:

- β_2-mediated vasodilation in skeletal muscle vascular beds *in the absence of α-mediated vasoconstriction*, along with vasodilation in renal and mesenteric vasculature → ↓ DBP → significant ↓TPR → ↓ MAP → reflex ↑ HR
- Reflex ↑HR plus + β_1-induced ↑ HR → tachycardia, which can lead to dysrhythmias

Check your knowledge

The chart illustrates the cardiovascular Drug X, which activates autonomic receptors. What are the most likely receptor affinities of Drug X?

A. α_1, α_2
B. α_1, α_2, β_1
C. β_1, β_2
D. M_2
E. N_M

Parameter	Control	Drug X
SBP	120 mm Hg	110 mm Hg
DBP	85 mm Hg	55 mm Hg
HR	60/min	120/min

Summary of Sympathomimetics, Part 1

- The sympathomimetics are grouped into subclasses related to their direct effects on adrenergic receptors, the indirect-acting sympathomimetics, mixed-acting agents.
- The structural variations of the drugs determine the relative affinities for adrenergic receptors, catecholamine transporters, and to degradation by monoamine oxidase and catechol-*O*-methyltransferase.
- The pharmacological properties of both the direct- and indirect-acting drugs relate to the classification, distribution, and mechanisms of the α and β receptors.
- Depending on their receptor specificities, the drugs mimic the physiologic effects characteristic of the catecholamines, including increasing heart rate and force of cardiac contraction, modifying systemic peripheral resistance, metabolic and endocrine actions, CNS actions on respiratory stimulation, wakefulness and psychomotor activity, and appetite reduction.
- Sympathomimetic vasopressors and inotropes are mainstays in the management of shock and heart failure.
- Adverse effects are extensions of their effects on α and β receptors: excessive vasoconstriction, cardiac arrhythmias, myocardial infarction, hemorrhagic stroke, pulmonary edema, and more.

Check your knowledge

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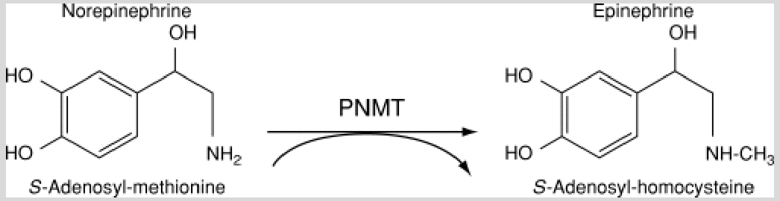
Where is EPI synthesized and secreted from (in the periphery)?

- EPI is synthesized in the adrenal medulla: dopamine → norepinephrine → epinephrine
- Phenylethanolamine-N-methyltransferase converts NE to EPI.
- PNMT is present in the adrenal medulla



What are EPI's relative affinities for adrenergic receptors?

- $\beta_1 = \beta_2 > \alpha_1 = \alpha_2$ EPI has strong affinity for all adrenergic receptors:

What are the effects of physiologic EPI on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

Acronyms: See slides 14, 15

- ↑ chronotropy and ↑ inotropy (β1) → rise in CO → ↑SBP
- vasodilation in skeletal muscle beds (β2) → ↓ peripheral vascular resistance → ↓DBP
- MAP is not greatly elevated (usually) → reflex actions that reduce HR are not triggered
- ↑HR, ↑SV, ↑CO and constriction of veins (venoconstriction) → ↑ venous return to the heart, which boosts SV and overall CO.
- NOTE: EPI in high dose or continuous infusion causes vasoconstriction and increased total peripheral resistance.

Check your knowledge

EPI is the drug of choice for what condition?

- Anaphylaxis / anaphylactic shock (hypotension due to widespread vasodilation and vascular leakage leading to reduced flow to organs and organ failure)

Why is it the drug of choice (actions and effects)?

- Anaphylactic shock is deadly. EPI works fast and prevents progression of symptoms.
- It causes bronchodilation ($\beta 2$ action) which opens airways
- It causes vasoconstriction ($\alpha 1$, $\alpha 2$) and increases the force of cardiac contraction($\beta 1$) which increase blood pressure.
- Vasoconstriction also reduces allergic angioedema (with throat swelling) and hives
- Suppresses release of histamine and other mediators from mast cells

Check your knowledge

Where is NE synthesized and secreted from?

- NE is an adrenergic neurotransmitter synthesized in presynaptic adrenergic neurons, stored in vesicles and released into the synapse where it activates adrenergic receptors at effector organs.
- dopamine $\rightarrow$ norepinephrine by dopamine-β-hydroxylase
- NE is also synthesized in the adrenal medulla and secreted into the bloodstream in relatively small amounts normally.

What are NE's relative affinities for adrenergic receptors?

- $\alpha_1 = \alpha_2 > \beta_1 >>> \beta_2$

What are the effects of intravenous NE on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

- Vasoconstriction $\rightarrow$ ↑TPR $\rightarrow$ ↑SBP and ↑DBP
- ↑ inotropy (β1) $\rightarrow$ ↑SV $\rightarrow$ modest ↑ CO
- ↑ MAP is elevated but compensatory vagal reflex overcomes the direct stimulation of HR (β1) actions $\rightarrow$ HR is unchanged or may decrease

Check your knowledge

NE is the drug of choice for what conditions?

- Circulatory shock: profound hypotension with low peripheral resistance
- septic shock, cardiogenic shock, neurogenic shock
- Drug selection is based on patient-specific factors.

Why is it the drug of choice (actions and effects)?

- Clinical experience of NE’s effectiveness.
- Potent vasoconstriction to increase systemic vascular resistance and raise blood pressure
- Mild increase in cardiac output compared to phenylephrine
- Lower risk of arrhythmias compared to dopamine and dobutamine
- NOTE: Vasopressors are components of a broader treatment plan.

Check your knowledge

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What are the dose-dependent effects of intravenous dopamine infusion on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

- Low dose: **D_1** activation in renal and mesenteric vessels → vasodilation →↑renal blood flow, ↑GFR, and ↑Na^+ excretion and ↓ peripheral vascular resistance
- Intermediate dose: **β_1 ARs** activation → ↑contractility (direct effect) + causes NE release from nerve terminals (indirect effect) → positive inotropic effect → ↑systolic blood pressure; diastolic pressure same or slightly increased and total peripheral resistance is usually unchanged
- High dose: **α_1 ARs** activation → general vascular constriction (pressor effect)

What are the effects of intravenous phenylephrine infusion on blood pressure, chronotropy, inotropy, and peripheral vascular resistance?

- Phenylephrine cause generalized vasoconstriction → ↑peripheral resistance, ↑SBP, and ↑DBP with marked reflex bradycardia and stroke volume may decrease (slowed HR + lack of positive inotropic effect)

What are the clinical indications for dopamine and phenylephrine?

- Both are alternatives for NE in the treatment of shock.
- Phenylephrine also has topical applications for reducing congestion by inducing local vasoconstriction.

Check your knowledge

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A. α_1, α_2
B. α_1, α_2, β_1
C. β_1, β_2
D. M_2
E. N_M

Parameter	Control	Drug X
SBP	120 mm Hg	110 mm Hg
DBP	85 mm Hg	55 mm Hg
HR	60/min	120/min

C. Isoproterenol

- The chart shows that Drug X has unopposed beta-1 and beta-2 action →↓SBP ↓DBP ↑HR
- A decrease in mean blood pressure, an increase in pulse pressure, plus a marked increase in heart rate are characteristic of a drug like isoproterenol.
- PVR and mean BP are decreased because of activation of beta-2 receptors in the vasculature.
- Systolic BP decreases less than diastolic BP because of activation of beta1 receptors in the heart, leading to an increase in stroke volume, as well as the increase in heart rate.

Resources

- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e 2023, Chapter 14: Adrenergic Agonists and Antagonists
- Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 9: Adrenergic Agonists & Sympathomimetics
- UpToDate Lexidrug monographs

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4RR